A Survey of Fake News Detection Using Machine Learning

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Journal of Demonstration Studies, 2021

Abstract

Fake news spreads quickly online and harms public trust. This survey reviews machine learning methods for detecting fake news, including support vector machines, random forests and BERT-based classifiers. We compare their reported accuracy and discuss open problems.

Keywords: fake news detection, machine learning, text classification, misinformation

1. Introduction

Online platforms allow misinformation to spread within hours. Automated detection of fake news is therefore an active research problem. This survey asks which machine learning methods detect fake news most reliably.

2. Related Work

Early work used hand-crafted linguistic features. Later work moved to deep neural networks and pre-trained language models.

3. Methodology

We reviewed papers published between 2015 and 2021. Each paper was categorised by method family: classical classifiers such as SVM and Random Forest, and deep models such as BERT. Reported accuracy was tabulated by dataset.

4. Results

BERT-based models reached up to 96.2 percent accuracy on benchmark datasets. SVM and Random Forest reached 88 to 92 percent. Deep models consistently outperformed classical classifiers.

5. Limitations

Deep models have a high computational cost. Most benchmark datasets are small and contain only English text. The survey could not compare studies that used different evaluation protocols.

6. Conclusion

Pre-trained language models are currently the strongest approach to fake news detection, at the price of computational cost.

7. Future Work

Future work should study multilingual fake news detection. Lightweight models that run on modest hardware should also be developed.

References

[1] J. Example, An Early Study of Linguistic Cues in Deceptive Text, Journal of Demonstration Studies, 2015.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.